

Artificial Intelligence, Machine Learning, & Robotics in Business

Restaurants in the Digital Age: Embrace Technology to Drive Efficiency and Customer Satisfaction

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Abstract: The integration of technology into the restaurant industry has transformed operations, customer experiences, and employee roles, especially following the COVID-19 pandemic. This evolution includes reliance on cloud-based systems, mobile applications, and AI-driven solutions, presenting both opportunities and challenges. Enhanced digital ordering, personalized experiences, and data-driven strategies have reshaped consumer expectations, while innovations like voice AI in drive-thrus and chatbots optimize efficiency. However, the dependency on uninterrupted internet connectivity heightens operational vulnerabilities. To adapt, industry leaders must embrace, champion, and effectively communicate technological advancements to both employees and customers. This proactive approach ensures streamlined operations and builds confidence in emerging technologies, fostering resilience in a rapidly evolving digital landscape.

Keywords: Restaurant Technology, Digital Ordering, Artificial Intelligence, Customer Experience, Internet Connectivity, Voice AI

In my role as an executive in the restaurant industry I am asked often – What keeps you up at night? For the last several years my answer has been the same one, losing the internet. My point of sales system (POS) will go down without access to the cloud, my credit cards will stop processing, mobile applications such as loyalty and order ahead will stop functioning and the back of house (BOH) computer system will lose critical functionality. In my career, there has always been a reliance on technology, but it was not always so integrated into the daily operations of the restaurant. BYOB has a much different meaning today than it did when I was in college. Now it stands for Bring Your Own Broadband as we ensure we are always connected. When we can't find a reliable provider, we utilize 3G/4G mobile backup systems or leveraging satellite technology which just elevates my anxiety that at any moment that unit will go down. While technological innovation has always been important in my 30-year career it has accelerated exponentially in the last few years. The Covid-19 Pandemic changed the way restaurants operated in 2020 and one of the primary results was increased use of online applications and technology to

reduce human contact. The technological innovations implemented during the Pandemic have changed consumer behavior and expectations in the quick-service and fast casual dining industry. Here is an example I utilize with my students in Tourism & Hospitality Business Perspectives class for the UF Online Master's Program in Tourism & Hospitality Management. What restaurant apps have you downloaded onto your phone and why? During the Pandemic this allowed you to reduce human contact by taking the menu interaction, order placement and money handling online. Now you will arrive at the restaurant and be able to pick up the prepared order without interacting with the restaurant staff. My family took this functionality and then incorporated it into our busy lifestyles for efficiency. We would search online for an available restaurant on our route and order ahead while on the way home. Now instead of joining a line and waiting for our order to be taken and completed we schedule the pickup in advance and do not wait. We call this out in our advertising to the consumer and encourage them to download the app, order ahead and "skip the line".

We also refer to a transaction coming in electronically on our mobile app to be a digital order. We then add our self-service kiosks, 3rd party delivery apps and online ordering systems to calculate total digital mix. This number has grown significantly in the last several years and has some restaurants approaching 50% of their total sales mix coming from these applications. So why has our industry embraced this technology? First, it is what the consumers have demonstrated that they want. Secondly, there are significant benefits to our business as well. Let me take you through several here:

1. **Convenience:** Mobile apps provide convenience and accessibility for customers to the business anytime and from anywhere they have their device. This streamlined experience can lead to increased customer satisfaction and loyalty if the app is easy to use.
2. **Personalization:** Mobile apps allow companies to gather data on customer preferences, behavior, and interactions within the app. This data can be used to personalize recommendations, offers, and content tailored to individual customers. This is also why it is important for the business to have ownership or access to the data. 3rd Party apps are not obligated to share their user data. Whoever controls the app, controls access to the data.
3. **Direct Communication:** Mobile apps enable direct communication channels between businesses and customers through push notifications, in-app messages, and alerts. Companies can use these communication tools to send targeted promotions, updates, and relevant information to customers.
4. **Enhanced Features:** Mobile apps often offer additional features and functionalities that may not be available through other channels, such as customized ordering options, loyalty program integration, mobile payments, geolocation services, and access to content. These features can enhance the user experience and differentiate the app from competitors. This is a big strategy the companies I have worked for have utilized to encourage downloads and have customers join the loyalty program.
5. **Data Insights:** Mobile apps generate valuable data and insights on user behavior, preferences, and interactions. The companies then use this data to make informed business decisions and set strategy.

The other exciting development impacting our restaurant industry is the increasing usage of artificial intelligence. For those on the front lines of the hospitality industry who want to understand how this technology is impacting the entire enterprise, I recommend the textbook from Dr. Rachel Fu, (2023) *Artificial Intelligence, Machine Learning, and Robot Applications in*

Hospitality Business. Kendal Hunt Publishing Company. AI enables machines to perform tasks that typically require human intelligence. This allows the machine to learn and adapt to the task it is being asked to perform as it gathers and processes more data and experiences.

Artificial Intelligence (AI) is being increasingly utilized in various aspects of the restaurant industry to enhance operations, improve customer experiences, and drive efficiency. Here are a few examples of where AI is impacting the restaurants I operate:

1. **Personalized Recommendations:** AI-powered systems analyze customer data, such as order history and preferences, to provide personalized menu recommendations and suggestions. This helps restaurants offer customized experiences and upsell relevant items to customers.
 - I mentioned earlier the importance of owning or having access to your customer data. You would not be able to provide personalized recommendations unless you had the guest's purchase history in which to build the recommendations from.
 - When a guest arrives at our restaurant they are asked if they have a personalized code to check in with. This allows the system to access their loyalty account. The system can provide suggestions based on purchase history as well as allow our team members to use the guest's name in a more personalized conversation.
2. **Chatbots and Virtual Assistants:** Restaurants are using AI-driven chatbots and virtual assistants to handle customer inquiries, reservations, and order processing.
 - Think of an app or an online site you utilize that has a chatbot pop up in the bottom corner and asks if you need any help. Organizations are using this technology to quickly respond to guest inquiries that do not necessarily need human interaction. Here is an example, you ordered through the app and after you picked up the order you receive a push message asking you how your experience was. Your order had a missing item or perhaps it was not customized to your request. You respond with a low score and let the app know why. The system will follow an established algorithm to recover you with a free item or other offer. The offer can now be tracked to not only know if you returned and when but will also track the amount you purchased as a result of the recovery matrix.
 - One of the most exciting uses of AI impacting my restaurants right now is the use of voice AI in my drive-thrus. Several of my restaurants have been testing the functionality of this technology and more importantly the willingness of our customers to interact with it. The system continues to evolve and learns from the interactions with the customers and the feedback the team provides the developers. For the customers who are not comfortable with the technology we can have a service team member jump in and take over. Some customers had concerns that the technology was eliminating human jobs but that was not the case. We have found that AI technology is a productivity enhancer. The team member can do another task while listening to the interaction on their headset and jump in to resolve any issue the AI is not capable of answering. The team members are embracing technology and giving the voice AI system a human name when referring to it. The management team embraces the AI system and refers to it as one of their best employees. They know it is not going to call in sick or miss work. However, that takes me to my original fear – the internet must be working, or I lose the voice AI system.

In conclusion, we are not going to go backwards in terms of technology and its' use in the restaurants, so I find it is very important to embrace the change. That means I need to champion

the new technology to my leaders and the team members, so they are willing to try it, as well as give them the tools to be successful with it. This also means that I must help them to be able to explain how the technology works with our customers that are not comfortable with it. With a little extra effort everyone will be able to become comfortable and proficient with it.



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